

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	08 July 2026
Team ID	
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Ingestion & Integration	Import EV charging station data into MySQL Import vehicle specification and battery sensor data Connect the MySQL data source to Tableau
FR-2	Data Preprocessing & Cleaning	Remove duplicate records and handle missing (NULL) values Split columns (e.g., CheapElectricCars) and rename fields (e.g., Brands) Change field data types (e.g., Address to Geographic Role) for map visualizations
FR-3	Dashboard & Visualization	Charging pattern & usage analysis dashboard Battery performance vs. driving range dashboard Comparative EV model efficiency dashboard
FR-4	Filtering & Interactivity	Filter by brand, region, price, and body style Filter by powertrain type and vehicle name Export dashboards and stories as PDF reports

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboards should be intuitive and easy to navigate for non-technical users, including city planners, fleet managers, and

		everyday EV consumers, with minimal training required.
NFR-2	Security	Access to charging, vehicle, and location data should be role-based and protected, ensuring only authorized users can view or export sensitive information.
NFR-3	Reliability	The system should consistently deliver accurate, up-to-date data and visualizations with minimal errors or downtime.
NFR-4	Performance	Dashboards should load and refresh within a few seconds, even when processing large datasets across multiple regions and vehicle brands.
NFR-5	Availability	The platform should be available 24/7 so city planners, fleet managers, and consumers can access insights whenever needed.
NFR-6	Scalability	The system should scale to support additional cities, EV brands, charging stations, and users without requiring a redesign of the core architecture.